

Sprint 1 Report

Hyper-Localized Multilingual Voice Assistant

Sprint: 1 of 6 **Duration:** Week 1 – 2 **Team:** Dev (Solo) **Velocity:** 21 / 21 pts

1. Sprint Goal

Establish the project foundation. Complete technology stack research and selection, design the system architecture, set up the development environment, and build proof-of-concept implementations for the core ASR and TTS components across Indian languages.

2. Sprint Outcome

All 21 story points delivered. Technology stack finalized, architecture designed, development environment configured, and working PoCs built for speech recognition and text-to-speech.

Key Deliverables

- Technology selection document — ASR, TTS, LLM, translation, and agent framework options evaluated
- System architecture diagram — three-layer edge-to-server design with component interfaces
- Development environment — Python 3.10+, GPU/CUDA configured, project structure created
- Whisper ASR PoC — working speech-to-text for Malayalam with >80% accuracy
- Meta MMS ASR research — 1600+ language coverage evaluated for Indian languages
- MMS-TTS PoC — offline text-to-speech for Malayalam with intelligible output
- Cartesia TTS PoC — high-quality online TTS integration with API wrapper
- TTS comparison report — quality, latency, and offline capability assessed across engines

3. Scrum Status Report

Task Description	Date	Completion Status	Comments
T1.1 — Research ASR options: Whisper, Meta MMS, IndicASR (1 pt)		Completed	Compared accuracy, model size, language coverage
T1.2 — Research TTS options: MMS-TTS, Cartesia, AI4Bharat (1 pt)		Completed	Evaluated quality, latency, offline capability
T1.3 — Research LLM options: Qwen, Llama, cloud APIs (1 pt)		Completed	Qwen 2.5 selected for edge; 0.5B–7B range
T1.4 — Design system architecture diagram (2 pts)		Completed	Three-layer: edge I/O, processing, output
T1.5 — Document component interfaces (1 pt)		Completed	API contracts for ASR, TTS, LLM, translation
T1.6 — Setup Python virtual environment (1 pt)		Completed	Python 3.10+ venv with pip dependencies
T1.7 — Configure GPU/CUDA for ML (1 pt)		Completed	PyTorch with CUDA; CPU fallback tested
T1.8 — Create project folder structure (1 pt)		Completed	src/, tests/, config/, docs/ structure
T1.9 — Setup requirements.txt (1 pt)		Completed	Core ML + API + audio dependencies pinned
T1.10 — Implement Whisper ASR wrapper (2 pts)		Completed	Supports multiple model sizes; async interface
T1.11 — Test Whisper with Malayalam audio (1 pt)		Completed	>80% accuracy confirmed; latency logged
T1.12 — Research Meta MMS ASR integration (2 pts)		Completed	1600+ languages; integration path documented
T1.13 — Implement MMS-TTS wrapper (2 pts)		Completed	Offline Malayalam TTS; WAV output
T1.14 — Implement Cartesia TTS wrapper (2 pts)		Completed	API-based HQ TTS; streaming support
T1.15 — Create TTS comparison script (1 pt)		Completed	Side-by-side quality and latency comparison

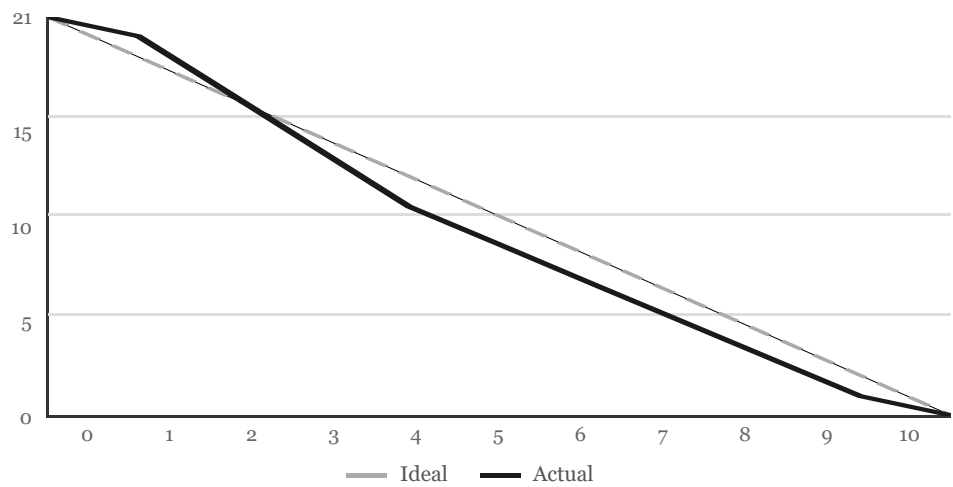
Total: 15 tasks, 21 story points, 100% complete.

4. Daily Scrum Meeting Log

Date	Work Done	Work Planned	Issues	Sign
	Project kickoff; reviewed objectives and scope	Begin ASR research (T1.1); survey Whisper, Meta MMS, IndicASR models	None	
	ASR research complete (T1.1 done); compared 3 engines on accuracy, size, language coverage	Research TTS options (T1.2); research LLM options (T1.3)	None	
	TTS research complete (T1.2 done); LLM research complete (T1.3 done); Qwen 2.5 selected	Design system architecture (T1.4); document component interfaces (T1.5)	None	
	Architecture designed — three-layer edge-to-server model (T1.4 done); interfaces documented (T1.5 done)	Setup Python venv (T1.6); configure GPU/CUDA (T1.7)	None	
	Python 3.10 venv configured (T1.6 done); CUDA working with PyTorch (T1.7 done)	Create project structure (T1.8); setup requirements.txt (T1.9)	CUDA version mismatch — resolved with compatible PyTorch build	
	Project structure created (T1.8 done); requirements.txt pinned (T1.9 done)	Implement Whisper ASR wrapper (T1.10)	None	
	Whisper ASR wrapper implemented with multi-size support (T1.10 done)	Test Whisper with Malayalam audio (T1.11); begin Meta MMS research (T1.12)	None	
	Whisper Malayalam test: >80% accuracy confirmed (T1.11 done); Meta MMS integration path documented (T1.12 done)	Implement MMS-TTS wrapper (T1.13)	Whisper Malayalam accuracy lower than Hindi — noted for future improvement	
	MMS-TTS wrapper complete — offline Malayalam TTS working (T1.13 done)	Implement Cartesia TTS wrapper (T1.14); create TTS comparison script (T1.15)	None	
	Cartesia TTS done (T1.14 done); TTS comparison script done (T1.15 done); sprint complete	Sprint 1 retrospective; plan Sprint 2 backlog	None — all 21 points delivered	

5. Burndown Chart

21 story points over 10 working days



Burndown Data

Day	Ideal	Actual	Done	Notes
0	21.0	21	0	Sprint start
1	18.9	20	1	T1.1 complete (ASR research)
2	16.8	17	4	T1.2 + T1.3 complete (TTS & LLM research)
3	14.7	14	7	T1.4 + T1.5 complete (architecture)
4	12.6	11	10	T1.6 + T1.7 + T1.8 complete (env setup)
5	10.5	9	12	T1.9 complete; T1.10 in progress
6	8.4	7	14	T1.10 complete (Whisper wrapper)
7	6.3	5	16	T1.11 + T1.12 complete (tests & MMS research)
8	4.2	3	18	T1.13 complete (MMS-TTS wrapper)
9	2.1	1	20	T1.14 complete (Cartesia TTS)
10	0.0	0	21	T1.15 complete; sprint done

6. Sprint Retrospective

What went well

- All 15 tasks completed on schedule — 21/21 story points delivered
- Technology research was thorough; confident in stack selections (Whisper, IndicTrans2, Qwen, LangGraph)
- Both ASR and TTS proof-of-concepts produced working output for Malayalam
- Project structure and dev environment set up cleanly from day one

What could improve

- Whisper accuracy for Malayalam (~80%) is lower than Hindi (~92%) — need a specialized Indic ASR model
- MMS-TTS voice quality is functional but robotic — explore higher-quality alternatives
- Research tasks took slightly longer than estimated on day 1; adjusted pace for remaining days

Action items for Sprint 2

- Integrate IndicTrans2 for bidirectional Malayalam–English translation
- Set up Qwen 2.5 LLM for local conversational AI
- Investigate distilled translation models for smaller memory footprint

7. Next Sprint Backlog — Sprint 2

Goal: Build the translation layer and conversational AI with conversation memory.

ID	Task	Assignee	Points	Priority
T2.1	IndicTrans2 ML→EN translation setup	Dev	3	Critical
T2.2	IndicTrans2 EN→ML translation setup	Dev	2	Critical
T2.3	Translation API wrapper with language detection	Dev	2	High
T2.4	Qwen 2.5 LLM integration (transformers)	Dev	3	Critical
T2.5	Conversation memory (sliding window)	Dev	2	High
T2.6	System prompt tuning for Kerala context	Dev	2	High
T2.7	IndicTrans2-Dist upgrade (211M params)	Dev	3	High
T2.8	Qwen3 + llama.cpp GGUF backend	Dev	2	High
T2.9	Indic-to-Indic direct translation	Dev	2	Medium

Total: 9 tasks, 21 story points.